

TOWARDS ZERO-SHOT ANOMALY DETECTION AND REASONING WITH MULTIMODAL LARGE LANGUAGE MODELS

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What ?

We introduce a comprehensive framework for explainable anomaly detection with three key contributions:

- **Anomaly-OneVision (Anomaly-OV):** specialist visual assistant for Zero-Shot AD & Reasoning
- **Look-Twice Feature Matching (LTFM)** mechanism to capture fine-grained details as human perception
- **Anomaly-Instruct-125k:** The first large-scale dataset for visual anomaly reasoning.

Why ?

- **Data Scarcity:** Traditional AD methods require massive normal data, which is impractical for new products
- **Lack of Explainability:** Existing ZSAD methods (e.g., WinCLIP) act as "black boxes," providing scores without reasons.
- **Limitations of Generalist MLLMs:** Powerful models like GPT-4o often fail to detect tiny, subtle defects due to a lack of specialized visual focus

Overview

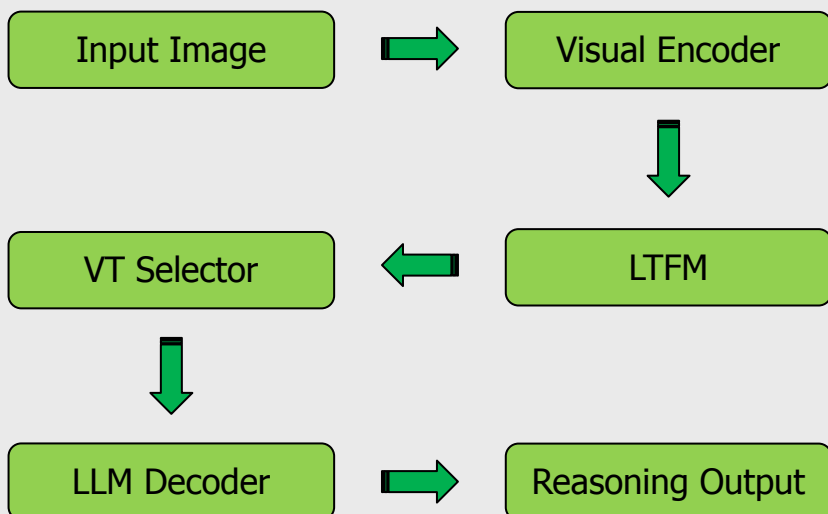


Diagram 1. Process flow of Anomaly-OV framework

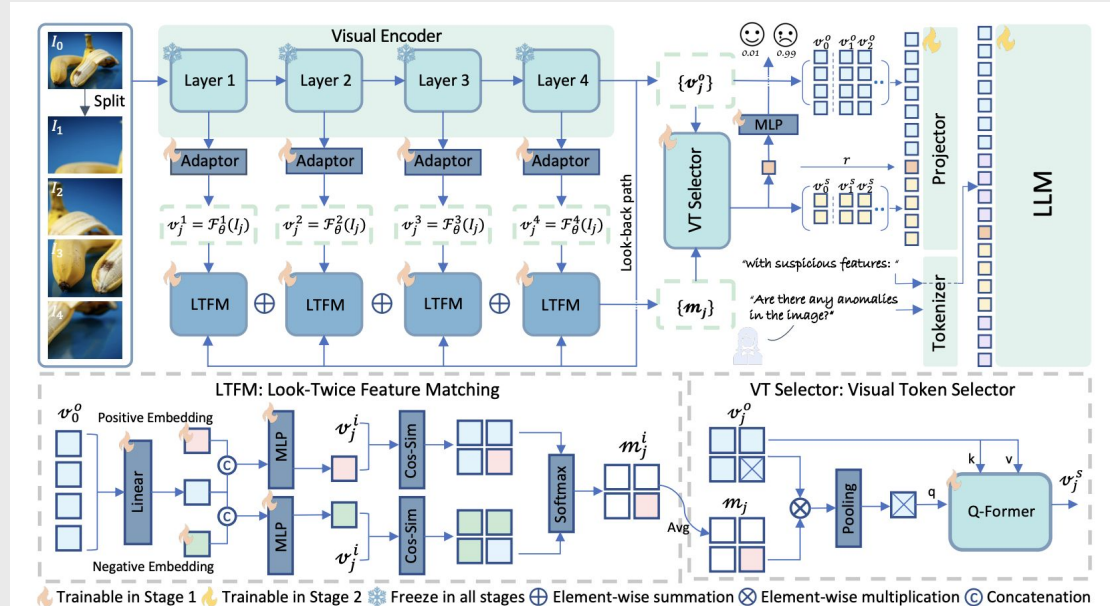


Figure 1. The Anomaly-OV overview framework integrates a specialized Anomaly Expert

Description

Methodology

Look-Twice Feature Matching (LTFM)

Inspired by human inspection behavior:

- **Look 1 (Global):** The model scans the whole image to understand the object context ("This is a candle").
- **Look 2 (Local):** It matches patch features against learned "Normal" and "Abnormal" embeddings to identify suspicious regions.

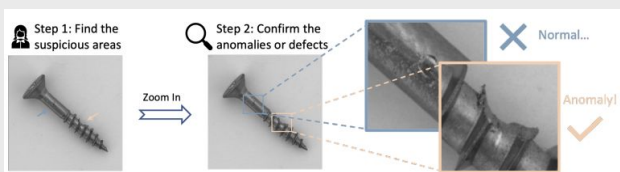


Figure 2. Simulation of visual anomaly inspection by humans

Visual Token Selector

Problem: High-resolution images produce thousands of tokens (noise).

Solution: Uses the Significance Map to filter out background tokens.

Result: Only "suspicious" tokens are zoomed in and sent to the LLM, reducing hallucination and improving focus.

Data Construction & Training

Anomaly-Instruct-125k Dataset

An automated pipeline using GPT-4o and Google Images to collect 72k "in-the-wild" anomaly images.

Using In-context Learning to generate 125k multi-turn conversations (*Detection* → *Description* → *Cause* → *Solution*).

Two-Stage Training Strategy

Stage 1 (Professional Training): Trains the Anomaly Expert (Projector & Adaptor) using Balanced BCE Loss to distinguish normal/abnormal features.

Stage 2 (Visual Instruction Tuning): Freezes the fine-tunes the LLM to enable natural language reasoning capabilities.

Experiments & Results

Zero-Shot Detection Performance (AUROC):

- MVTEC AD: **94.0%** (Outperforming WinCLIP & AnomalyCLIP)
- VisA: **91.1%** (+5.3% improvement)

Reasoning Capabilities:

- Anomaly-OV achieves **79% accuracy** in text-based detection, surpassing GPT-4o (70%)
- Qualitative Comparison: Anomaly-OV demonstrates **more precise** detection of fine-grained details than GPT-4o

Model	Anomaly Detection				Low-level Reasoning			Complex Reasoning	
	Accuracy	Precision	Recall	F1-score	ROUGE-L	SBERT	GPT-Score	SBERT	GPT-Score
GPT-4V [71]	0.68	0.90	0.49	0.55	0.16	0.65	3.31	0.77	5.64
GPT-4o [72]	0.70	0.83	0.71	0.68	0.24	0.71	4.84	0.81	6.89
Qwen2-VL-2B [87]	0.65	0.87	0.55	0.59	0.22	0.55	1.94	0.74	4.26
Qwen2-VL-7B [87]	0.76	0.91	0.69	0.75	0.25	0.61	3.09	0.68	4.62
InternVL-2-8B [13]	0.74	0.78	0.81	0.76	0.23	0.73	3.69	0.80	5.08
InternVL-2-26B [13]	0.73	0.86	0.66	0.68	0.21	0.74	4.13	0.80	5.49
IXC-2.5-7B [101]	0.72	0.88	0.63	0.67	0.21	0.58	2.45	0.77	5.14
LLaVA-OV-0.5B [44]	0.54	0.70	0.19	0.28	0.20	0.63	2.54	0.81	4.34
LLaVA-OV-7B [44]	0.71	0.95	0.56	0.63	0.24	0.66	3.57	0.79	5.44
LLaVA-OV-0.5B*	0.71	0.77	0.84	0.76	0.31	0.70	3.69	0.82	5.31
Anomaly-OV-0.5B	0.79	0.86	0.83	0.82	0.33	0.72	3.87	0.83	5.67
Anomaly-OV-7B	0.79	0.83	0.86	0.83	0.34	0.73	4.26	0.84	6.34

Figure 3. Anomaly-AD presents better AD & Reasoning performance compared with generalist MLLMs and fine-tuned base model (LLaVA-OV)